

Rakuten / **RakutenAI-7B**   like 51  Follow  Rakuten 295

 Text Generation  Transformers  PyTorch  Safetensors  English

 Japanese  mistral  text-generation-inference  arxiv:2403.15484

 License: apache-2.0

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 Model card  Files  xet

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 Safetensors 

Model size 7.37B params Tensor type BF16  Files info

 Inference Providers **NEW**

 Text Generation

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 Model tree for Rakuten/RakutenAI-7B

Base model	mistralai/Mistral-7B-v0.1
 Finetuned (992)	this model
Finetunes	5 models
Merges	1 model
Quantizations	7 models

 **RakutenAI-7B**

 **Model Description**

RakutenAI-7B is a systematic initiative that brings the latest technologies to the world of Japanese LLMs. RakutenAI-7B achieves the best scores on the Japanese language understanding benchmarks while maintaining a competitive performance on the English test sets among similar models such as

OpenCalm, Elyza, Youri, Nekomata and Swallow. RakutenAI-7B leverages the Mistral model architecture and is based on Mistral-7B-v0.1 pre-trained checkpoint, exemplifying a successful retrofitting of the pre-trained model weights. Moreover, we extend Mistral's vocabulary from 32k to 48k to offer a better character-per-token rate for Japanese.

The technical report can be accessed at [arXiv](#).

If you are looking for an instruction-tuned model, check [RakutenAI-7B-instruct](#).

If you are looking for a chat-tuned model, check [RakutenAI-7B-chat](#).

 Model Evaluation Results

Model Name	7-Avg. excl. XLSum-ja	Avg.	JCS	JNLI	MARC-ja	JSQuAD	Jaqket v2	XLSum-ja	xWino	M
			accuracy	accuracy	accuracy	exact-match	exact-match	rouge-2	accuracy	acc
			3-shots	3-shots	3-shots	2-shots	1-shot	1-shot	0-shot	5-
rakuten-ai-7b	69.80	62.83	84.27	48.69	96.29	79.09	80.67	14.08	77.16	2
nekomata-7b	66.01	58.83	85.43	40.14	96.80	76.29	71.99	8.59	73.83	1
japanese-stablelm-base-gamma-7b	64.83	59.12	80.07	14.71	92.41	81.38	85.05	19.16	82.59	1
youri-7b	62.71	56.90	76.94	51.11	90.96	57.45	78.09	16.27	78.00	
swallow-7b	60.86	55.18	78.91	15.16	90.27	73.28	80.24	15.41	76.96	1

Model Name	7-Avg. excl. XLSum-ja	Avg.	JCS	JNLI	MARC-ja	JSQuAD	Jaquet v2	XLSum-ja	xWino	M
elyza-japanese-Llama-2-7b	60.24	53.26	75.60	50.74	87.51	71.48	57.56	4.40	71.22	
elyza-japanese-Llama-2-7b-fast	58.31	51.34	71.49	45.77	86.61	70.91	64.18	2.54	61.63	
open-calm-7b	45.27	39.67	62.65	31.92	85.37	38.05	33.42	0.45	65.07	

Table1: RakutenAI-7B foundation model performance on Japanese LM-Harness metrics in comparison with other models.

Our model achieves the highest average score, more than 3 points ahead of the next best model. The models are sorted by 7-Avg. We use the following commit <https://github.com/Stability-AI/lm-evaluation-harness/tree/0fa86429679f521161d5b81a94c0c385e0a0976d> for Japanese LM-Harness with v0.3 prompt version.

Model Name	Avg.	ARC	HellaSwag	MMLU	TruthfulQA
		accuracy	accuracy	accuracy	accuracy
		25-shots	10-shots	5-shots	6-shots
rakuten-ai-7b	60.50	60.24	82.20	61.31	38.25
japanese-stablelm-base-gamma-7b	56.08	50.60	77.43	54.99	41.30
elyza-japanese-Llama-2-7b	52.76	51.62	76.54	44.85	38.02

Model Name	Avg.	ARC	HellaSwag	MMLU	TruthfulQA
elyza-japanese-Llama-2-7b-fast	52.07	51.79	75.46	44.41	36.63
nekomata-7b	51.97	47.35	72.78	48.38	39.38
youri-7b	50.60	49.15	75.02	42.36	35.89
swallow-7b	49.90	47.35	72.20	39.36	40.68
open-calm-7b	29.87	20.56	31.01	23.73	44.16

Table2: RakutenAI-7B foundation model performance on English LM-Harness metrics in comparison with other models.

Our model achieves the highest average score, more than 4 points ahead of the next best model. We use the following commit for English LM-Harness <https://github.com/EleutherAI/lm-evaluation-harness/tree/b281b0921b636bc36ad05c0b0b0763bd6dd43463>.

An independent evaluation by Kamata et.al. for [Nejumi LLMリーダーボード Neo](#) using a weighted average of [llm-jp-eval](#) and [Japanese MT-bench](#) also confirms the highest performance of chat/instruct versions of RakutenAI-7B among Open LLMs of similar sizes, with a score of 0.393/0.331 respectively, as of 22nd March 2024.

 Usage

```
from transformers import AutoModelForCausalLM, AutoTokenizer

model_path = "Rakuten/RakutenAI-7B"
tokenizer = AutoTokenizer.from_pretrained(model_path)
model = AutoModelForCausalLM.from_pretrained(model_path, torch_dtype="auto", device_map="auto")
model.eval()

requests = [
    "南硫黄島原生自然環境保全地域は、自然",
    "The capybara is a giant cavy rodent",
]
```

```

for req in requests:
    input_ids = tokenizer.encode(req, return_tensors="pt").to(device=model.device)
    tokens = model.generate(
        input_ids,
        max_new_tokens=256,
        do_sample=True,
        repetition_penalty=1.1,
        pad_token_id=tokenizer.eos_token_id,
    )
    out = tokenizer.decode(tokens[0], skip_special_tokens=True)
    print("INPUT:\n" + req)
    print("OUTPUT:\n" + out)
    print()
    print()

```

🔗 Model Details

- **Developed by:** [Rakuten Group, Inc.](#)
- **Language(s):** Japanese, English
- **License:** This model is licensed under [Apache License, Version 2.0](#).

🔗 Limitations and Bias

The suite of RakutenAI-7B models is capable of generating human-like text on a wide range of topics. However, like all LLMs, they have limitations and can produce biased, inaccurate, or unsafe outputs. Please exercise caution and judgement while interacting with them.

🔗 Citation

For citing our work on the suite of RakutenAI-7B models, please use:

```

@misc{rakutengroup2024rakutenai7b,
  title={RakutenAI-7B: Extending Large Language Models for Japanese},
  author={{Rakuten Group, Inc.} and Aaron Levine and Connie Huang and Chenguar},
  year={2024},

```

```
eprint={2403.15484},  
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}
```